

Service Authorization Policy

In Meican

Introduction

- Check caller identity
- Protect your app inner-cluster
- mTLS

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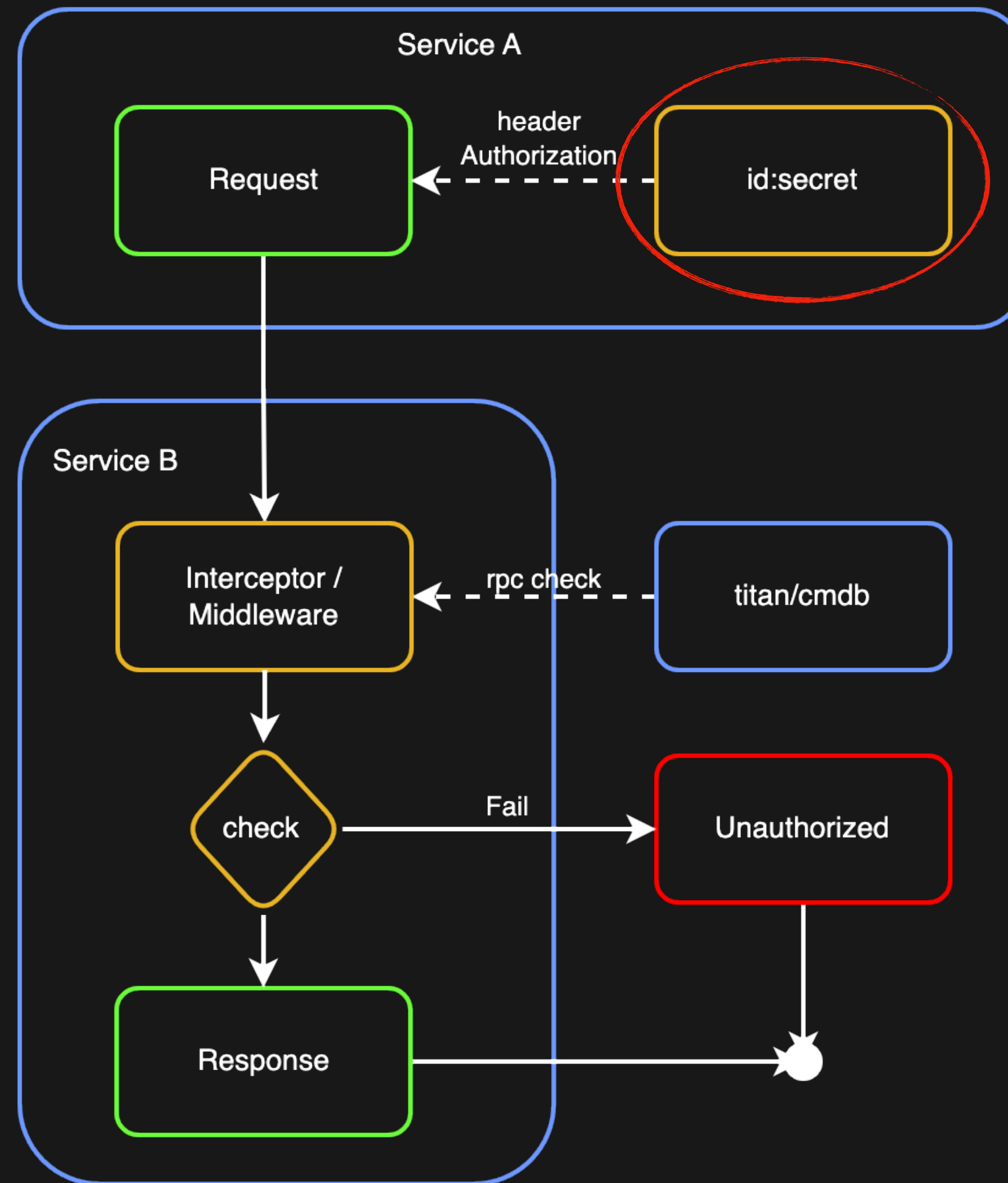
Basic Auth

Authorization: Basic base64(username:password)

Basic Auth

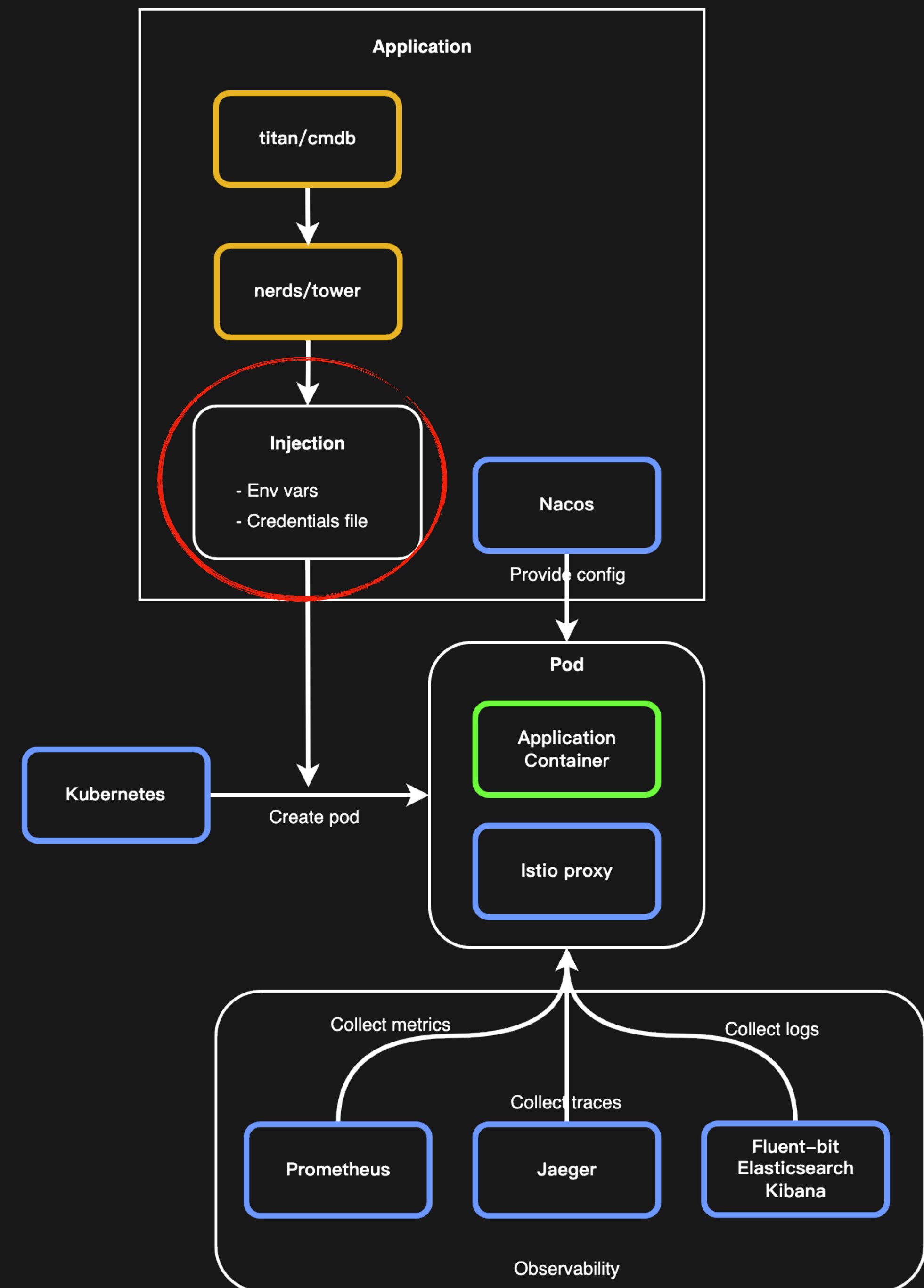
- Used in titan projects
- Additional development
 - Interceptor / Middleware
 - RPC calls
- Pretty straightforward

Basic Auth Activity Diagram



Basic Auth

Where the id:secret comes from ?



Basic Auth

- File mounted to container's filesystem
- Env variable: APP_CREDENTIALS
- By default: /var/run/secrets/tower/tower_resource_schema.toml

```
kind = "resource-schema"
updateTime = "2022-02-17T03:07:51.8693152Z"

[identifier]
errCodeAncestor = 100000
group = "app-titan"
privateKey = "-----BEGIN RSA PRIVATE KEY-----\nMIIEE"
project = "cmdb"
subset = "server"
publicKey = "-----BEGIN PUBLIC KEY-----\nMIIBCgKCAQI"
secret = "HIDE_FOR_SECURITY"
slug = "app-titan-cmdb-server"
uniqueID = "3efe222f-6d60-4c22-bb9e-2bc2e19b8b3b"
```

Basic Auth

- Definition: <https://gitlab.planetmeican.com/nerds/app/-/blob/master/pkg/credentials/credentials.go#L36>

```
// Identifier ...
type Identifier struct {
    // ErrCodeAncestor is generated via nerds/app-cli command apply
    ErrCodeAncestor int64 `toml:"errCodeAncestor"`

    // Slug is the unique identifier of your application, which could be considered
    // as the "username" of your application
    Slug string `toml:"slug"`

    // Group of your application
    Group string `toml:"group"`

    // Project of your application
    Project string `toml:"project"`

    // Subset of that application.
    // This is optional unless you want to have multiple subsets under the same project.
    // This value will potentially affect the Slug value.
    // Each Subset should have different secret and key pairs, but always inherit one
    // same ErrCodeAncestor.
    Subset *string `toml:"subset"`

    // UniqueID that associated to every application subset.
    // This shall be used with Secret to perform basic auth
    UniqueID string `toml:"uniqueID"`

    // Secret could be considered as the "password" of your application. You will need
    // Slug and Secret to exchange credentials from Tower
    Secret string `toml:"secret"`

    // KeyPair is a dedicated rsa 2048 key pair that generated via nerds/app-cli command apply.
    // Usually you can exchange your public key with other system in order to generate signature
    // to secure your communication.
    KeyPair
```

```
/app # curl -H 'Authorization: Basic YTpi' http://cmdb-server.app-titan.svc.cluster.local:8024/v1/teams/titan &&
{"code":16, "message":"WithinTx.fn: rpc error: code = Unauthenticated desc = basic auth failed: application subse
/app # echo YTpi'' | base64 -d && echo ''
a:b
```



```
/app # curl -H 'Authorization: Basic Yjk1ZmU0MDEtMjMwZi00YTZlLWI1NDAtZjEyM2Q3NGE1MDljOjBUMDA5UHEyYkJ4cTB2Yw==' http://cmdb-server.app-titan.svc.cluster.local:8024/v1/teams/titan &&
{"team":{"name":"titan", "times":{"createTime":"2022-03-07T10:29:52.697580Z", "updateTime":"2022-03-07T10:29:52.697580Z"}}, "code":0, "message":""}
/app # echo 'Yjk1ZmU0MDEtMjMwZi00YTZlLWI1NDAtZjEyM2Q3NGE1MDljOjBUMDA5UHEyYkJ4cTB2Yw==' | base64 -d && echo
b95fe401-230f-4a6e-b540-f123d74a509c:0T009Pq2bBxq0vc
```

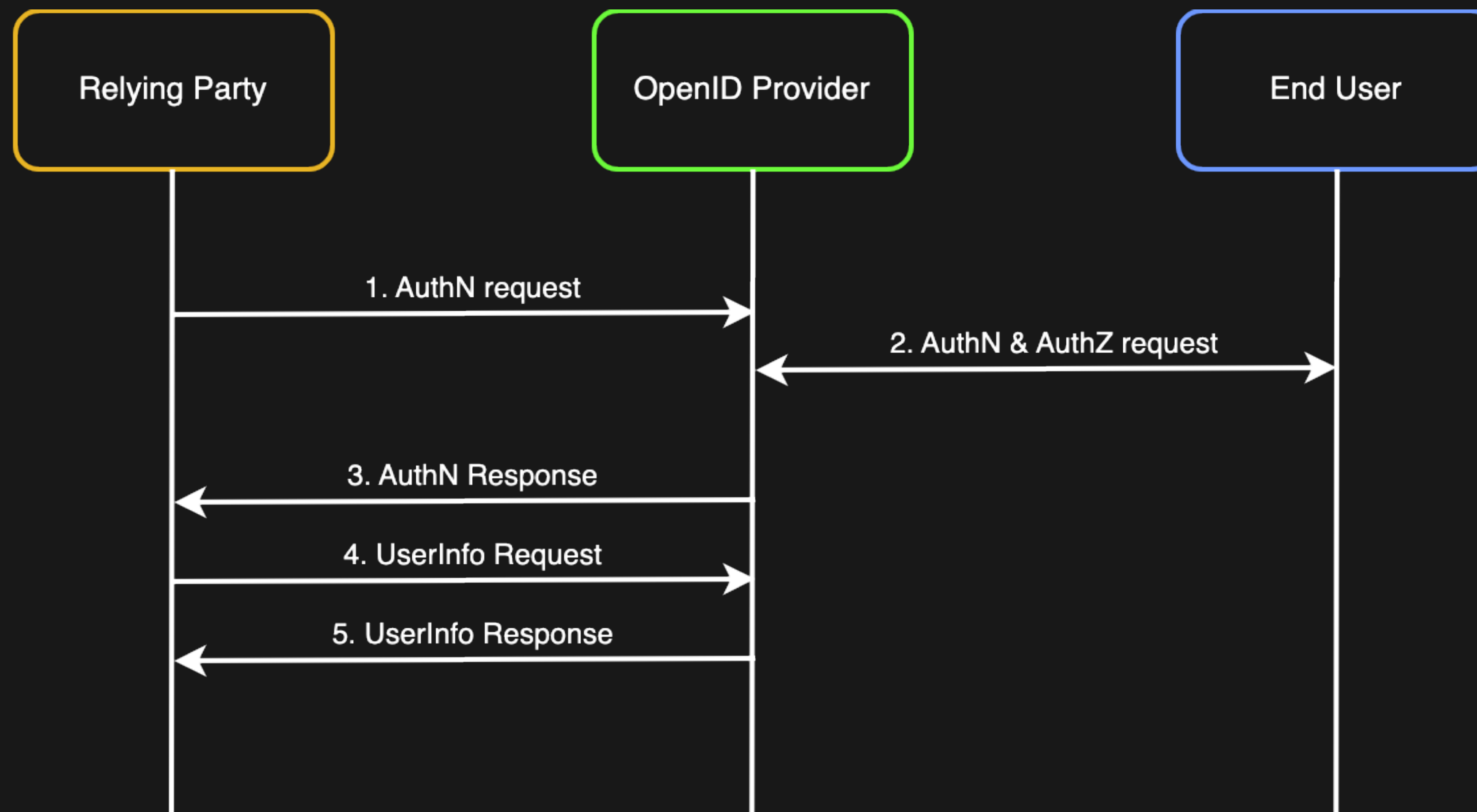


Basic Auth

Example

OpenID Connect

- OIDC
- Used in SSO (Apollo), Istio, AWS+Kubernetes, meican.x/authorizer
- Glossary:
 - JWKS: JSON Web Key Set
 - JWT: JSON Web Token

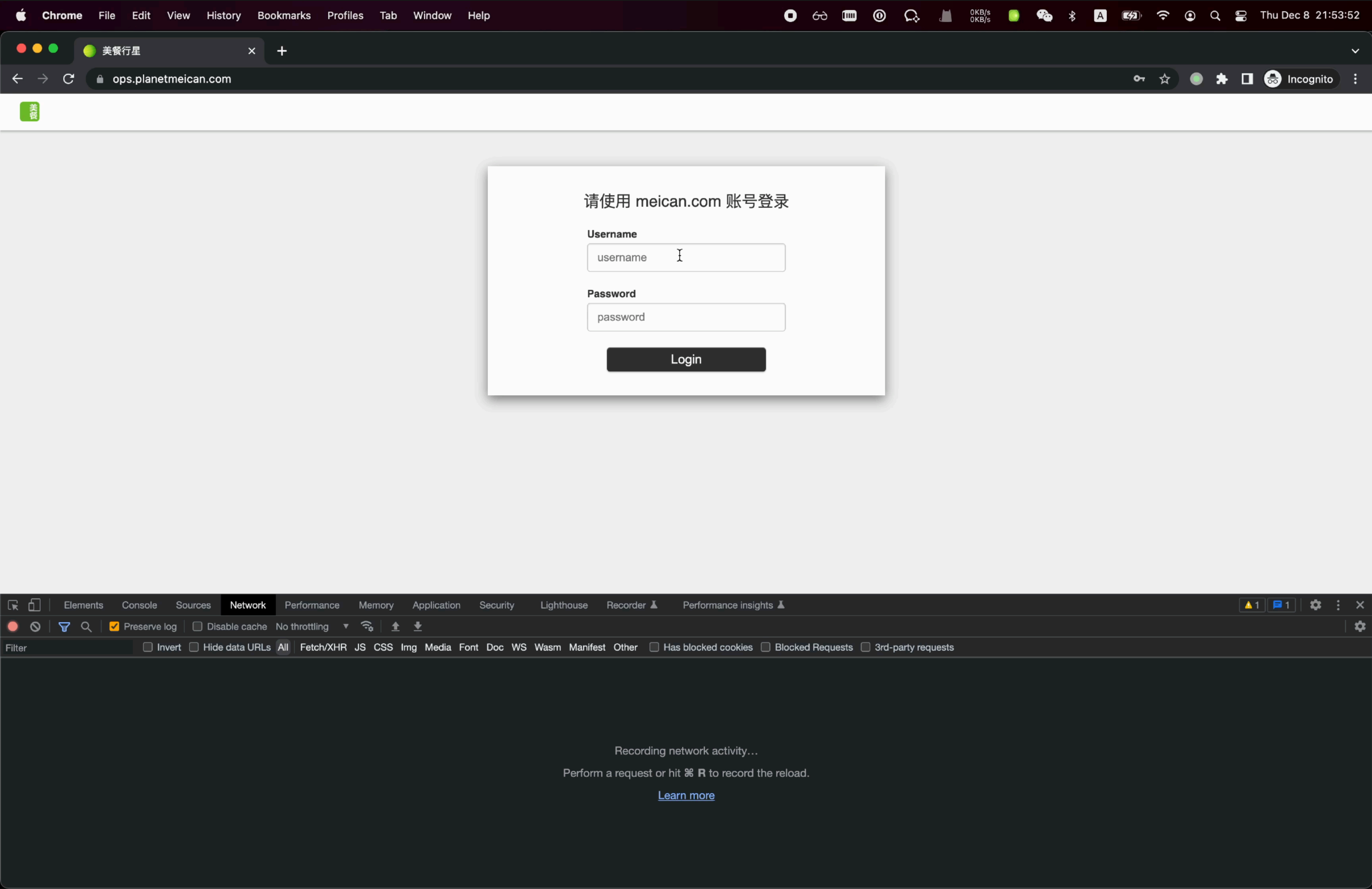


OIDC

Workflow

AuthZ & AuthN

- AuthZ: Authorization
 - Authorization determines access rights
- AuthN: Authentication
 - Verifies the identity of a user or service



OIDC

Live Demo



Encoded

PASTE A TOKEN HERE

```
eyJhbGciOiJSUzI1NiIsImtpZCI6ImduZGhiNXp
xWUY3LUtVWjY3ZVRvX1VEN3JmNXJKd3o2VXRUUn
ZzakRkRFkiLCJ0eXAiOiJKV1QiLCJ0eXN0eXB1Ij
E2Njg0ODMwNDcsImZlcnlucyI6InNlcnZlci5jbW
RiLnRpdGFuQGE5cm5sLWVrcy1mYW4ubWVpY2FuI
iwiOiIjoiY2xpZW50LmVzLnR5bWVpY2FuIn0uYm
VulPZE3_x5BQPbL0Jw0uQLugPB8iQ29-
ASRh1uQNNrbxz2ESrLUGnV4VJW2LwhXBt5f2-
yQL3w-h4qIDAVcISJbHitMrL-
Z6XvfBT0JxaShLN1r4-
Plscv_j0qE2WAsiuE_K2V77cbTnxP0yvgxvxeaH
JmmyoCU-
AbcRUl_ZTGXqb87tukLnF0mMuFVcEAL0qhVqr2b
1ZEK5zqCNevbGf8rs5yB3-
Il6MAZakkHmrHYn8Q7pUXGcnT0_NU1rsctL4jiN
Q6YkZG0rWn25hsuClWut1bnv0_IKHb5gesZWHaq
iW0BUAVy9fYduq15jJ-TVAGXKvyIhtvcazJ-
3Ra9g
```

Decoded

EDIT THE PAYLOAD AND SECRET

HEADER: ALGORITHM & TOKEN TYPE

```
{
  "alg": "RS256",
  "kid": "gndhb5zqYF7-KUZ67eTo_UD7rf5rJwz6UtTRvsjDdDY",
  "typ": "JWT"
}
```

PAYLOAD: DATA

```
{
  "iat": 1668483047,
  "iss": "server.cmdb.titan@ntrnl-eks-fan.meican",
  "sub": "client.istio-jwt-demo.nerds@ntrnl-eks-
fan.meican"
}
```

VERIFY SIGNATURE

```
RSASHA256(
  base64UrlEncode(header) + "." +
  base64UrlEncode(payload),
  Public Key in SPKI, PKCS #1,
  X.509 Certificate, or JWK string
  format.
```

JWT

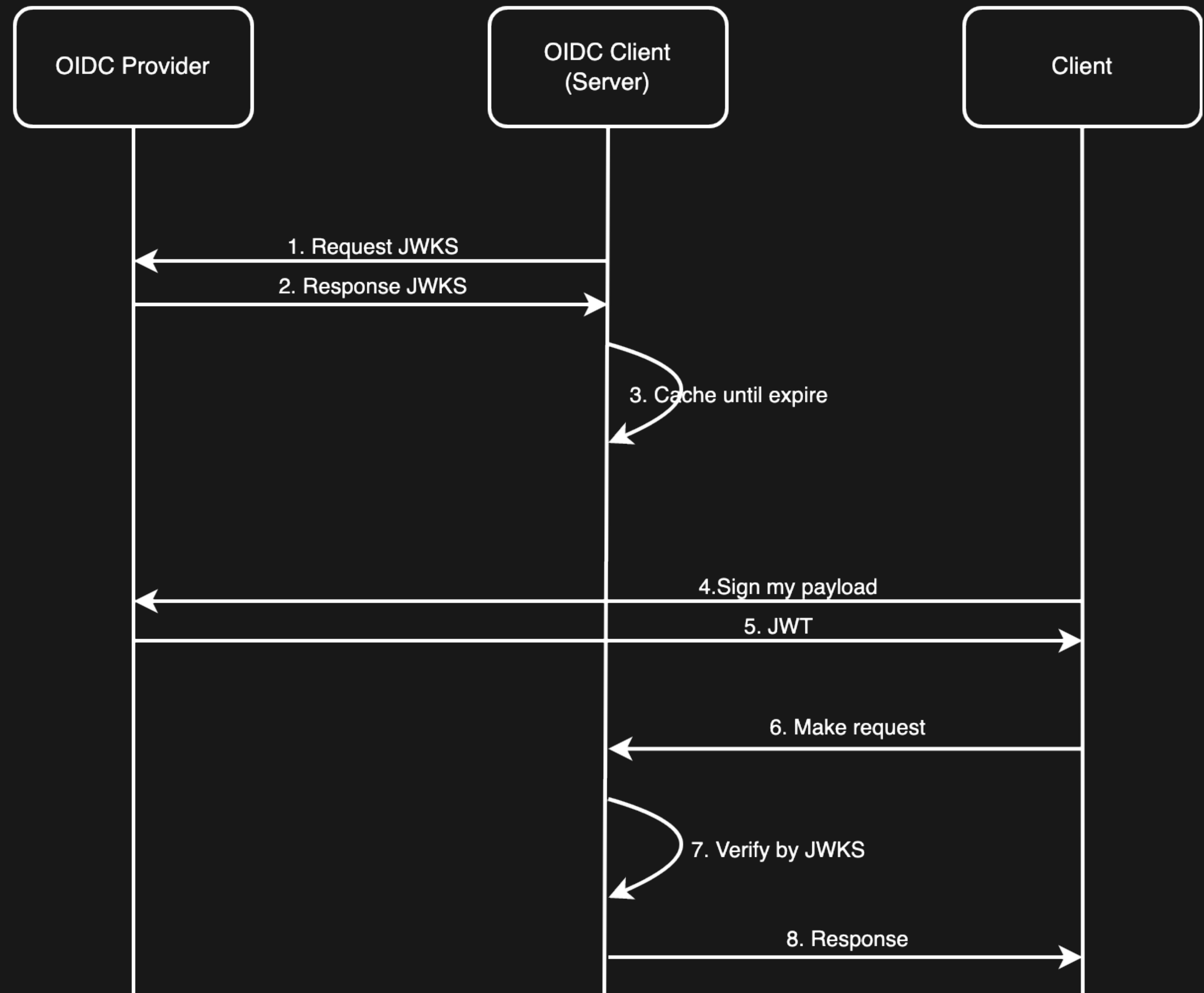
<https://jwt.io/>

```
{
  "keys": [
    {
      "use": "sig",
      "kty": "RSA",
      "kid": "b2141e833a1ac0718ea116d429c9a17a5e305fbe",
      "alg": "RS256",
      "n": "1qAbJvYhLikhJcZbSRafYYRys0WnITb-5whIC7XN0oB90p0wxXbnPsVfzjghWbuYS4Y4yHxEMPr-aXPdwm0uoH1ek225jgihCnNYV2WpcTtr",
      "e": "AQAB"
    },
    {
      "use": "sig",
      "kty": "RSA",
      "kid": "7822d935f4788107a29cb8ebb7b6a72fcb88e357",
      "alg": "RS256",
      "n": "q6TrDG6WGZJSqPP0GMh2dPCnmNrpjz6kpexN7hSqv0be3e1DtPH-3MTKtZwnYYWSVD17JCCoAsSxWwioAsmvE-AXP60_kVv0gTvJ0a_mcVgi",
      "e": "AQAB"
    }
  ]
}
```

JWKS

<https://www.rfc-editor.org/rfc/rfc7517>

OIDC JWKS - JWT



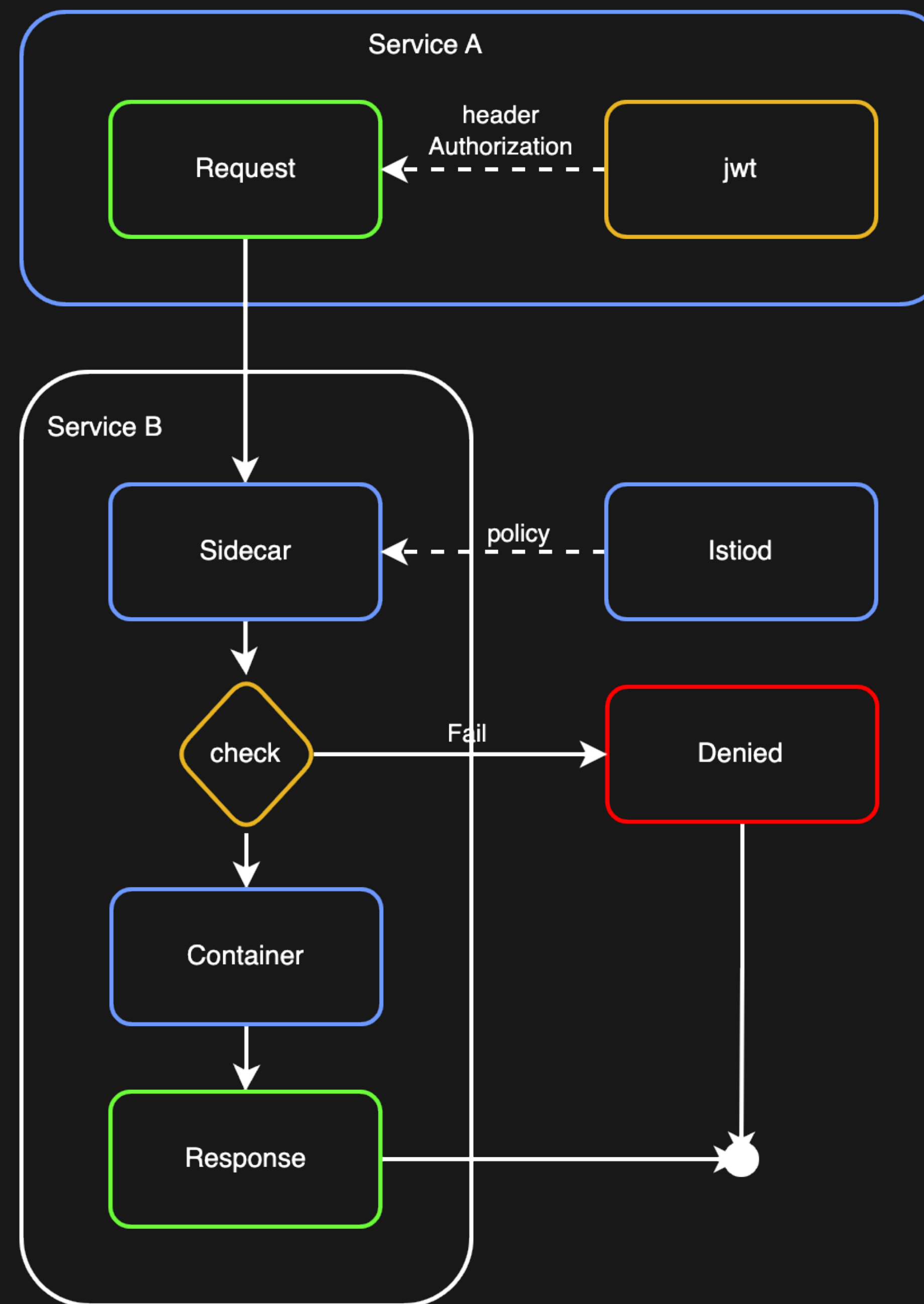
Istio Authz

- Service mesh - Implicit enablement
- No additional development
- No rpc call
- OpenID Connect

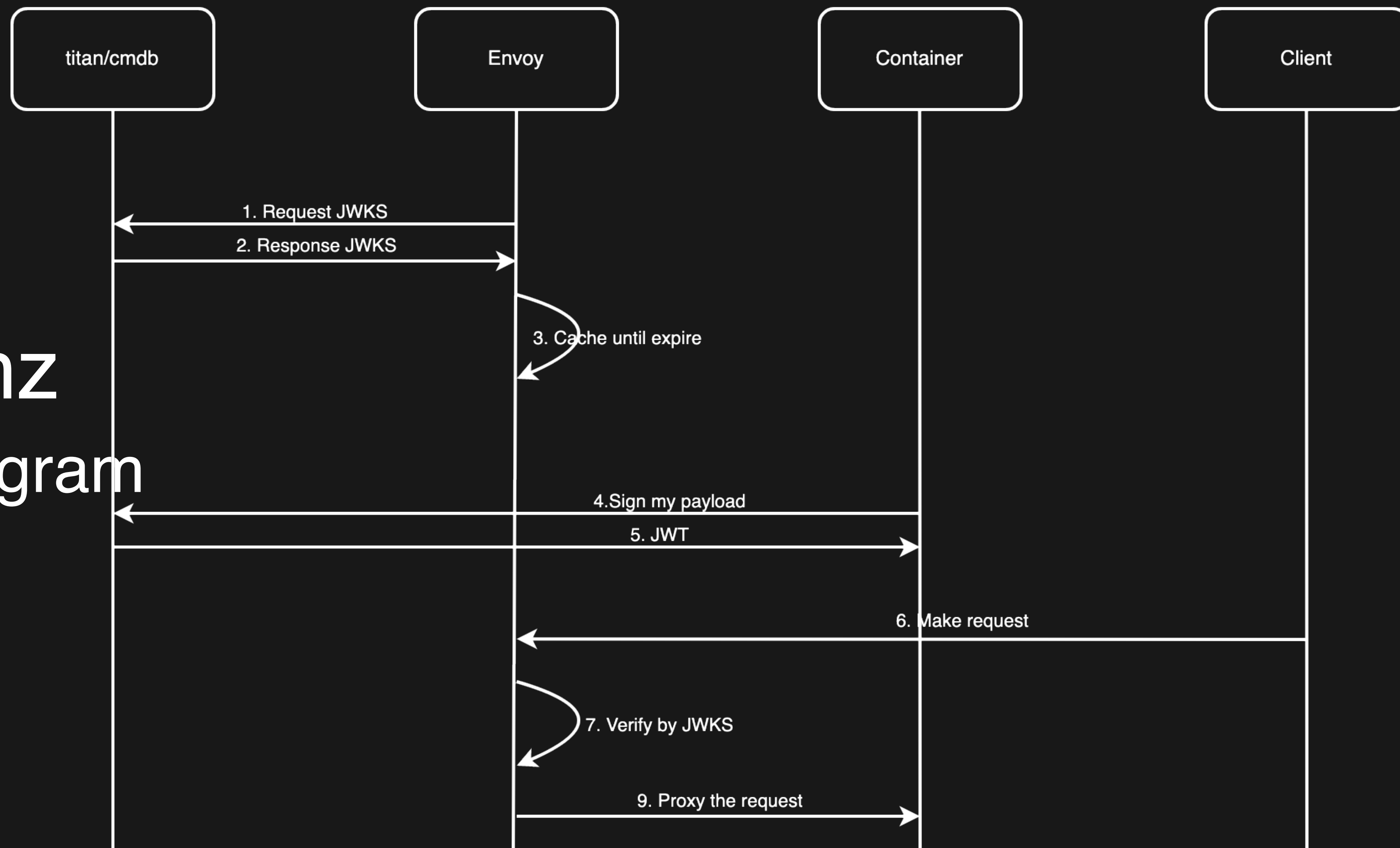
Istio Authz

Activity Diagram

<https://istio.io/latest/docs/concepts/security/#authorization>

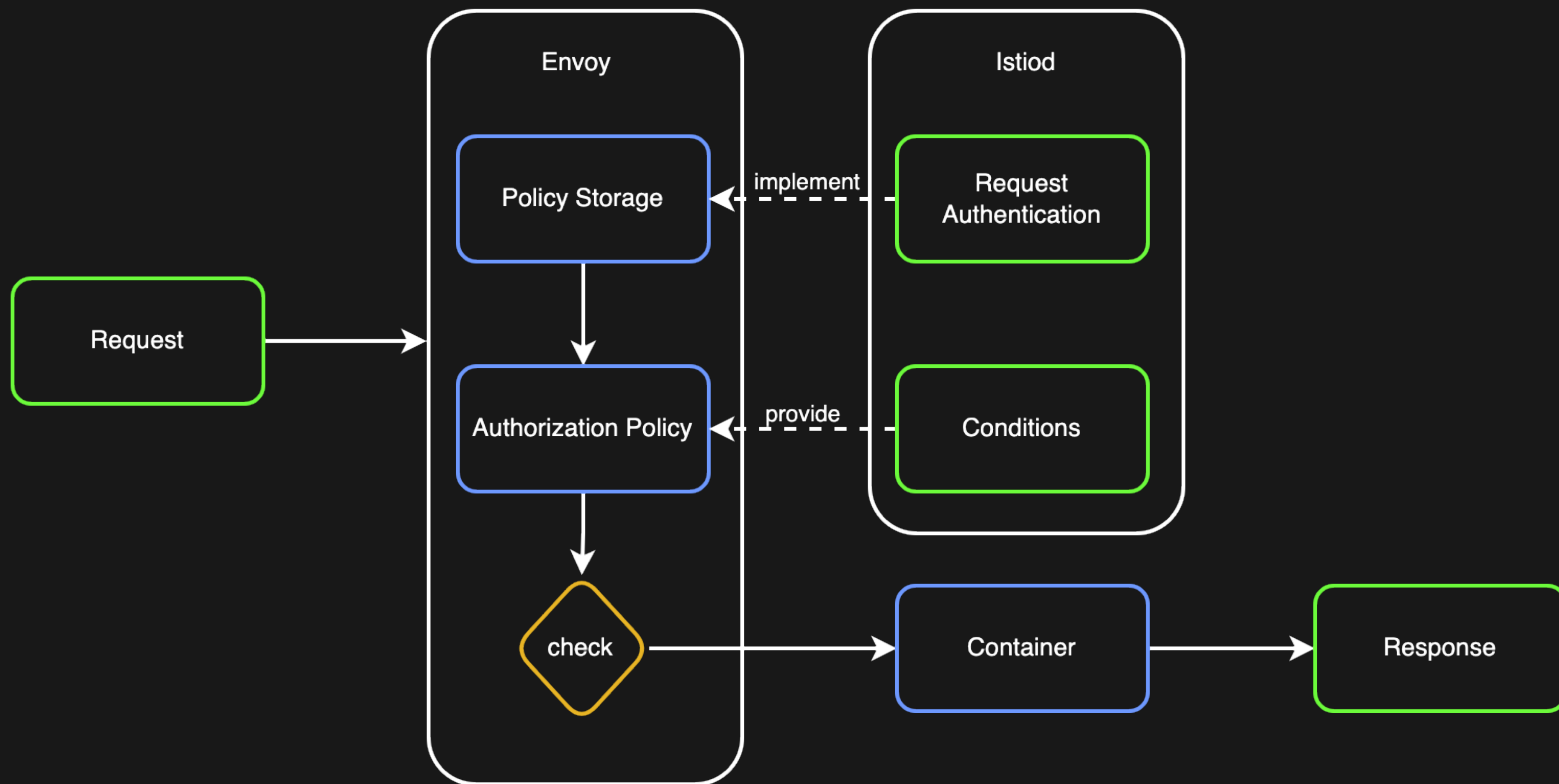


Istio Authz Sequence Diagram



Istio Authz

- Use OpenID Connect
- RequestAuthentication: defines what request authentication methods are supported by a workload
- AuthorizationPolicy: enables access control on workloads in the mesh.



Istio Authz

Dive into the policy

```

1  apiVersion: security.istio.io/v1beta1
2  kind: RequestAuthentication
3  metadata:
4    name: istio-jwt-demo-server-req-auth
5    namespace: app-nerds
6  spec:
7    selector:
8      matchLabels:
9        app: istio-jwt-demo
10       subset: server
11    jwtRules:
12      - issuer: "server.cmdb.titan@ntrnl-eks-fan.meican"
13        jwksUri: "http://cmdb-server-grpcgateway.app-titan.ntrnl-eks-fan.meican/oidc/keys"

```

Istio Request Authentication

Policy Storage

Workload selector

- Match pods to apply the RequestAuthentication & AuthorizationPolicy

```
spec:
  selector:
    matchLabels:
      app: istio-jwt-demo
      subset: server
```

JWT Rules

- JWT must be issued by:
 - server.cmdb.titan@ntrnl-eks-fan.meican
- JWKS retrieved from:
 - <http://cmdb-server-grpcgateway.app-titan.ntrnl-eks-fan.meican/oidc/keys>

```

1  apiVersion: security.istio.io/v1beta1
2  kind: AuthorizationPolicy
3  metadata:
4    name: istio-jwt-demo-server-auth-policy
5    namespace: app-nerds
6  spec:
7    selector:
8      matchLabels:
9        app: istio-jwt-demo
10     subset: server
11    action: ALLOW
12    rules:
13      - from:
14        - source:
15          requestPrincipals: [ "server.cmdb.titan@ntrnl-eks-fan.meican/client.istio-jwt-demo.nerds@ntrnl-eks-fan.meican" ]
16        - to:
17          operation:
18            ports: [ "18888" ]

```

Istio Authorization Policy

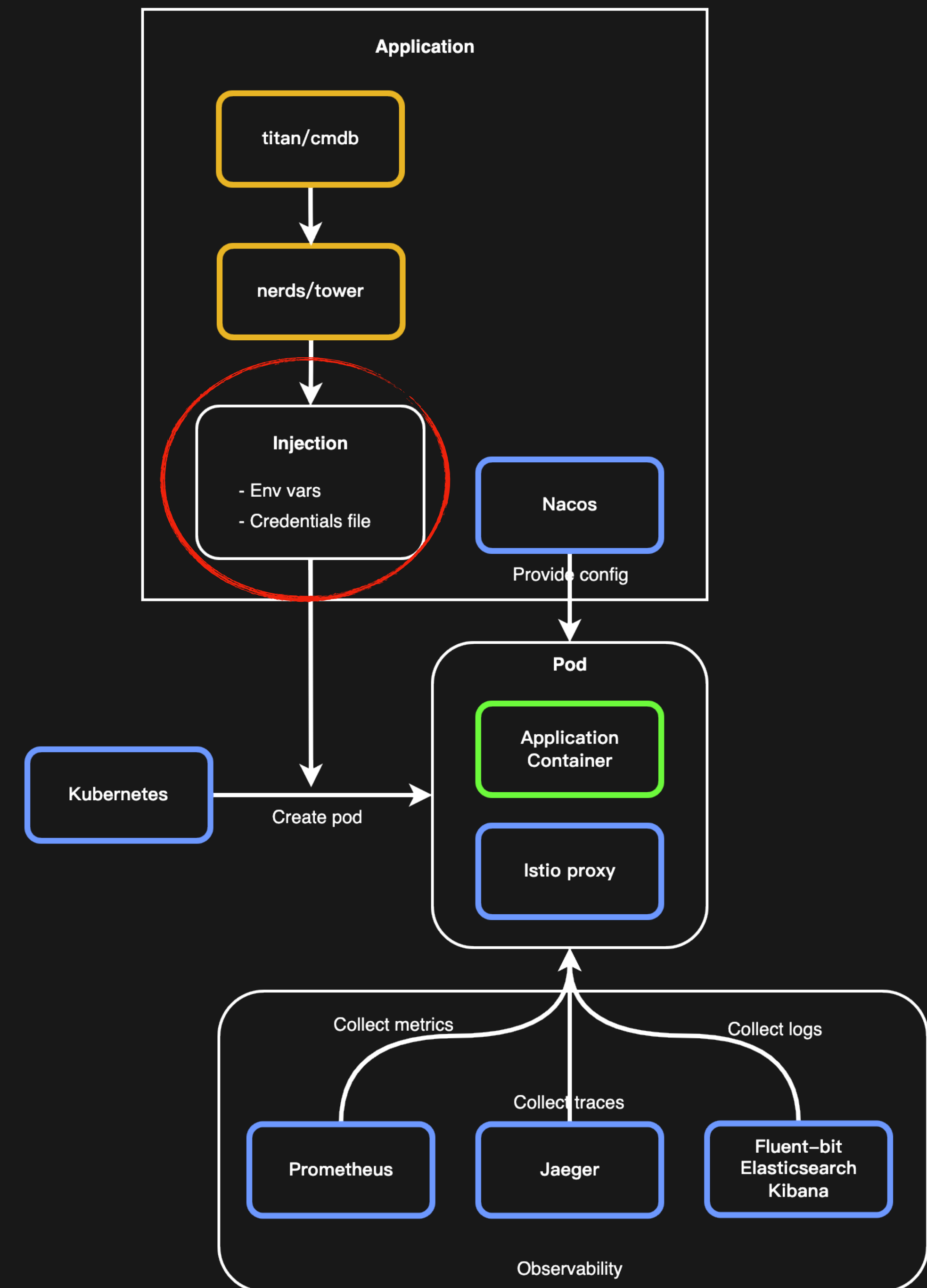
Conditions & Rules

Istio AuthorizationPolicy

- RequestPrincipals:
 - Issuer: server.cmdb.titan@ntrnl-eks-fan.meican
 - Subject: client.istio-jwt-demo.nerds@ntrnl-eks-fan.meican
- Exclude port 18888 for metrics collection

Istio Authz

Where the jwt comes from ?



Application JWT

- Definition: <https://gitlab.planetmeican.com/nerds/app/-/blob/master/pkg/credentials/credentials.go#L36>
- Version $\geq$ v0.10.6

```
// This value will potentially affect the Slug value.
// Each Subset should have different secret and key pairs, but always inherit one
// same ErrCodeAncestor.
Subset *string `toml:"subset"`

// UniqueID that associated to every application subset.
// This shall be used with Secret to perform basic auth
UniqueID string `toml:"uniqueID"`

// Secret could be considered as the "password" of your application. You will need
// Slug and Secret to exchange credentials from Tower
Secret string `toml:"secret"`

// KeyPair is a dedicated rsa 2048 key pair that generated via nerds/app-cli command apply.
// Usually you can exchange your public key with other system in order to generate signature
// to secure your communication.
KeyPair

// JWT is assigned to every registered application in order to do inter-app authorization.
// The subject of such jwt follows the following rules:
// - assuming the environment is sandbox, domain will be `ntrnl-eks-fan.meican`
// - if subset is undefined, the subject is `[project].[group]@[domain]`
// - if subset is defined, the subject is `[subset].[project].[group]@[domain]`
// Issuer usually is `server.cmdb.titan@[domain]` (domain is `ntrnl-eks-fan2.meican` when it
// See https://planetmeican.atlassian.net/browse/00PS-1176 for more information.
JWT string `toml:"jwt"`
}
```

Best practices

1. Read the doc: <https://planetmeican.atlassian.net/wiki/spaces/DEV/pages/473301014>
2. Apply ArgoCD Ephemeral Metadata: <https://argoproj.github.io/argo-rollouts/features/ephemeral-metadata/>
3. Apply Istio RequestAuthentication for canary deployment
 1. This will only deny requests with wrong JWT
 2. Requests without JWT still passes
4. Check the request with Authorization Header
5. Apply Istio AuthorizationPolicy for stable deployment
6. Check the request with Authorization Header

ArgoCD Ephemeral Metadata

- Adds labels to Pod for different stage
- Very useful for canary the workload selector
- Backlog: <https://planetmeican.atlassian.net/browse/OOPS-1249>

```
spec:
  strategy:
    canary:
      stableMetadata:
        labels:
          stage: stable
      canaryMetadata:
        labels:
          stage: canary
```

Add to ArgoCD/Rollout strategy

```
spec:
  selector:
    matchLabels:
      app: istio-jwt-demo
      subset: server
      stage: canary
```

Add to Istio CRDs selector

```
/app #
/app # curl -v -X POST http://istio-jwt-demo.app-nerds.svc.cluster.local:8024/ping && echo ''
* Trying 10.100.232.213:8024...
* Connected to istio-jwt-demo.app-nerds.svc.cluster.local (10.100.232.213) port 8024 (#0)
> POST /ping HTTP/1.1
> Host: istio-jwt-demo.app-nerds.svc.cluster.local:8024
> User-Agent: curl/7.79.1
> Accept: */*
>
* Mark bundle as not supporting multiuse
< HTTP/1.1 200 OK
< content-type: application/json
< grpc-metadata-content-type: application/grpc
< date: Mon, 05 Dec 2022 11:38:09 GMT
< content-length: 11
< x-envoy-upstream-service-time: 0
< server: istio-envoy
< x-envoy-decorator-operation: istio-jwt-demo.app-nerds.svc.cluster.local:8024/*
<
* Connection #0 to host istio-jwt-demo.app-nerds.svc.cluster.local left intact
{"data":""}
/app #
```

No Authorization Header



```
/app #
/app # curl -v -X POST -H 'Authorization: Bearer wrong.jwt.token' http://istio-jwt-demo.app-nerds.svc.cluster.local:8024/ping && echo ''
* Trying 10.100.232.213:8024...
* Connected to istio-jwt-demo.app-nerds.svc.cluster.local (10.100.232.213) port 8024 (#0)
> POST /ping HTTP/1.1
> Host: istio-jwt-demo.app-nerds.svc.cluster.local:8024
> User-Agent: curl/7.79.1
> Accept: */*
> Authorization: Bearer wrong.jwt.token
>
* Mark bundle as not supporting multiuse
< HTTP/1.1 401 Unauthorized
< www-authenticate: Bearer realm="http://istio-jwt-demo.app-nerds.svc.cluster.local:8024/ping", error="invalid token"
< content-length: 42
< content-type: text/plain
< date: Mon, 05 Dec 2022 11:40:47 GMT
< server: istio-envoy
< x-envoy-decorator-operation: istio-jwt-demo.app-nerds.svc.cluster.local:8024/*
<
* Connection #0 to host istio-jwt-demo.app-nerds.svc.cluster.local left intact
Jwt header is an invalid Base64url encoded
/app #
```

Incorrect Authorization Header



Istio Authz

Without AuthorizationPolicy (only RequestAuthentication)

```
/app # export TOKEN=eyJhbGciOiJSUzI1NiIsImtpZCI6ImduZGhiNXpxWUY3LUtVWjY3ZVRvX1VEN3JmNXJKd3o2VXRUNnZzakRkRFkiLCJ0eXAiOiJKV1QiQ.eyJpYXQiOiBwZXIyZW50LmlldGlWLWp3dC1kZW1vbm5lcmRzQG50cm5sLVVrcy1mYW4ubWVpY2FuIn0.cmVuelpZE3_x5BPblOJw0uQLugPB8iQ29-ASRh1uQNnrxbxz2ESrLUGnV4VV77cbTnxP0yvngxveaHJmmyoCU-AbcRUl_ZTGxqb87tukLnFomMuFvcEAL0qhVqr2b1ZEK5zaCNevbGf8rs5yB3-Il6MAZakkHmrHYn8Q7pUXGcnTO_NU1rsctL4jinQ6YkZGOrw
```

```
/app # curl -v -X POST -H "Authorization: Bearer $TOKEN" http://istio-jwt-demo.app-nerds.svc.cluster.local:8024/ping && echo ''  
* Trying 10.100.232.213:8024...  
* Connected to istio-jwt-demo.app-nerds.svc.cluster.local (10.100.232.213) port 8024 (#0)  
> POST /ping HTTP/1.1  
> Host: istio-jwt-demo.app-nerds.svc.cluster.local:8024  
> User-Agent: curl/7.79.1  
> Accept: */*  
> Authorization: Bearer eyJhbGciOiJSUzI1NiIsImtpZCI6ImduZGhiNXpxWUY3LUtVWjY3ZVRvX1VEN3JmNXJKd3o2VXRUNnZzakRkRFkiLCJ0eXAiOiJKV1QiQ.eyJpYXQiOiBwZXIyZW50LmlldGlWLWp3dC1kZW1vbm5lcmRzQG50cm5sLVVrcy1mYW4ubWVpY2FuIn0.cmVuelpZE3_x5BPblOJw0uQLugPB8iQ29-ASRh1uQNnrxbxz2ESrLUCE_K2V77cbTnxP0yvngxveaHJmmyoCU-AbcRUl_ZTGxqb87tukLnFomMuFvcEAL0qhVqr2b1ZEK5zaCNevbGf8rs5yB3-Il6MAZakkHmrHYn8Q7pUXGcnTO_NU1rsctL4jinQ6YkZGOrw  
>  
* Mark bundle as not supporting multiuse  
< HTTP/1.1 200 OK  
< content-type: application/json  
< grpc-metadata-content-type: application/grpc  
< date: Mon, 05 Dec 2022 11:44:48 GMT  
< content-length: 11  
< x-envoy-upstream-service-time: 0  
< server: istio-envoy  
< x-envoy-decorator-operation: istio-jwt-demo.app-nerds.svc.cluster.local:8024/*  
<  
* Connection #0 to host istio-jwt-demo.app-nerds.svc.cluster.local left intact  
{"data":""}
```

Correct Authorization Header



Istio Authz

Without AuthorizationPolicy (only RequestAuthentication)

```
/app #
/app # curl -v -X POST http://istio-jwt-demo.app-nerds.svc.cluster.local:8024/ping && echo
* Trying 10.100.232.213:8024...
* Connected to istio-jwt-demo.app-nerds.svc.cluster.local (10.100.232.213) port 8024 (#0)
> POST /ping HTTP/1.1
> Host: istio-jwt-demo.app-nerds.svc.cluster.local:8024
> User-Agent: curl/7.79.1
> Accept: */*
>
* Mark bundle as not supporting multiuse
< HTTP/1.1 403 Forbidden
< content-length: 19
< content-type: text/plain
< date: Mon, 05 Dec 2022 13:33:46 GMT
< server: istio-envoy
< x-envoy-decorator-operation: istio-jwt-demo.app-nerds.svc.cluster.local:8024/*
<
* Connection #0 to host istio-jwt-demo.app-nerds.svc.cluster.local left intact
RBAC: access denied
/app #
```

No Authorization Header



```
/app # curl -v -X POST -H "Authorization: Bearer $TOKEN" http://istio-jwt-demo.app-nerds.svc.cluster.local:8024/
* Trying 10.100.232.213:8024...
* Connected to istio-jwt-demo.app-nerds.svc.cluster.local (10.100.232.213) port 8024 (#0)
> POST /ping HTTP/1.1
> Host: istio-jwt-demo.app-nerds.svc.cluster.local:8024
> User-Agent: curl/7.79.1
> Accept: */*
> Authorization: Bearer eyJhbGciOiJSUzI1NiIsImtpZCI6ImduZGhiNXpxWUY3LUtVWjY3ZVRvX1VEN3JmNXJKd3o2VXRUUnZzakRkRFkiFuIiwic3ViIjoieY2xpZW50LmlzZGlvdWp3dC1kZW1vLm5lcmRzQG50cm5sLWVrcy1mYW4ubWVpY2FuIn0.cmVulpZE3_x5BQPbLOJw0uQLugPB8iE_K2V77cbTnxP0yvgxvxeaHJmmyoCU-AbcRUL_ZTGXqb87tukLnF0mMuFVcEAL0qhVqr2b1ZEK5zqCNeVbGf8rs5yB3-I16MAZakkHmrHYn8Q7pl
>
* Mark bundle as not supporting multiuse
< HTTP/1.1 200 OK
< content-type: application/json
< grpc-metadata-content-type: application/grpc
< date: Mon, 05 Dec 2022 11:49:03 GMT
< content-length: 11
< x-envoy-upstream-service-time: 0
< server: istio-envoy
< x-envoy-decorator-operation: istio-jwt-demo.app-nerds.svc.cluster.local:8024/*
<
* Connection #0 to host istio-jwt-demo.app-nerds.svc.cluster.local left intact
{"data":""}
/app #
```

Correct Authorization Header



Istio Authz

With AuthorizationPolicy

More about KeyPair

- RSA 2048 private/public keys
- PKCS #1 converted for Java
- Source: <https://gitlab.planetmeican.com/nerds/bazaar/-/blob/master/xsign/rsa.go>

```
// Slug is the unique identifier of your application, which could be considered
// as the "username" of your application
— Slug string `toml:"slug"`

// Group of your application
— Group string `toml:"group"`

// Project of your application
— Project string `toml:"project"`

// Subset of that application.
// This is optional unless you want to have multiple subsets under the same project.
// This value will potentially affect the Slug value.
// Each Subset should have different secret and key pairs, but always inherit one
// same ErrCodeAncestor.
— Subset *string `toml:"subset"`

// UniqueID that associated to every application subset.
// This shall be used with Secret to perform basic auth
— UniqueID string `toml:"uniqueID"`

// Secret could be considered as the "password" of your application. You will need
// Slug and Secret to exchange credentials from Tower
— Secret string `toml:"secret"`

// KeyPair is a dedicated rsa 2048 key pair that generated via nerds/app-cli command apply.
// Usually you can exchange your public key with other system in order to generate signature
// to secure your communication.
— KeyPair

// JWT is assigned to every registered application in order to do inter-app authorization.
// The subject of such jwt follows the following rules:
// - assuming the environment is sandbox, domain will be `ntrnl-eks-fan.meican`
// - if subset is undefined, the subject is `[project].[group]@[domain]`
```

More about KeyPair

- Exchange public key from titan/cmdb via gRPC/HTTP
- Signing your request's payload

```
-----BEGIN PUBLIC KEY-----
MIIBCgKCAQEA5M13f7uvSDwQMP6UW+MR+539Q5EVB61EsbJafAdtzgxQCGbBkXc1
nJxmyxgS99ere1LIWdIOa9/lPv/EiaWyow7JESldBzXbga8z150y5ByqHr0f2FX1
M9WvfTiWs5SB7oxBLVhWEUyapT0+hfA4zY6+i+gjXsgV++o7ju4dHrLhhxm0B1
76fZud0Yx9HdI340zB2hYcjv6KNMSon0e7vbxtspSiILJ7b9NmFQ4V6Kh54SQS21
ON5ZPI0mBLH4MgcowArL6MyICGQZPUxhlz13jn9dBz0fykmVK93K0L0oYX7LJ0/1
pFNU9bXeJSazf2ErcmjhFdQ06UT9F6PtwIDAQAB
-----END PUBLIC KEY-----
```

```
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEA5M13f7uvSDwQMP6UW+MR+539Q5EVB61EsbJafAdtzgxQCGb1
kXcpnJxmyxgS99ere1LIWdIOa9/lPv/EiaWyow7JESldBzXbga8z150y5ByqHr01
2FXjM9WvfTiWs5SB7oxBLVhWEUyapT0+hfA4zY6+i+gjXsgV++o7ju4dHrLhhx1
m0Bu76fZud0Yx9HdI340zB2hYcjv6KNMSon0e7vbxtspSiILJ7b9NmFQ4V6Kh541
QS2DON5ZPI0mBLH4MgcowArL6MyICGQZPUxhlz13jn9dBz0fykmVK93K0L0oYX71
J0/4pFNU9bXeJSazf2ErcmjhFdQ06UT9F6PtwIDAQABaoIBADxCELLHT0aK6wi1
Q35fsyC9XzIoR42aCHL0K3h2zGhzvbA4v8n7xC3G0xux32z+k98K0LE1js7D7LB1
1BAcX4uqHTIz3phDiugAqHqkn/6LvHlP5I0pMYk5QdqMyMPK3N1ietfiZ+i0H281
eehL0TUF10qEuJgKUV5Xc/oqKE4keXG3btrv5/MV7gwJlhGW1/+vM079HLfb1ST1
l21a0PuVMkPTVCA2B+RDgSM0DhlgRMtaJpm1Kax69z2Q6kMXrAokishTrobk/901
b76qx5I8XzxTY0BDG0iAk3vrXpU0waFloHJZRV6wih1yua+cq0qh9N0LvS6RK8s1
Cm0kFWECaYEA895ceYFwa0bSx/oKNCSHuW8vnN+VC1CeG8vacUa0BF4/REbR0101
```

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Fin.
Question ?